CHE 230: Molecular Systems in Chemical Engineering Homework 1

Due: 02/11/25

1. Graphite and Diamond are chemically identical but different. Explain why and give 2 other examples of materials that exhibit a similar phenomenon. (4 points)

- Diamond vs. Graphite
  - Same element (carbon), different atomic bonding.
  - Diamond: 3D tetrahedral covalent network → very hard, insulating.
  - Graphite: layered hexagonal structure → soft, electrically conductive.
- Iron (Fe): Ferrite vs. Austenite
  - Same element, different crystal structures.
  - Ferrite (BCC): magnetic, softer.
  - Austenite (FCC): non-magnetic, more ductile.
- Calcium Carbonate ($CaCO_3$): Calcite vs. Aragonite
  - Same chemical formula, different crystal structures.
  - Different density, stability, and mechanical properties.

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2. What is the main difference between ionic, covalent, and metallic bonds? Discuss and draw a diagram to explain. (6 points)

   The main difference is how electrons are distributed, either transferred, shared, or freely moving.

- Ionic bonds: Electrons are transferred between atoms, forming oppositely charged ions held together by electrostatic attraction.
- Covalent bonds: Electrons are shared between atoms, forming molecules or network structures.
- Metallic bonds: Valence electrons are delocalized and shared among many atoms, forming a conductive lattice

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3. Aluminum forms an FCC structure and has an atomic radius of 0.143 nm (9 points)
   a. Calculate the atomic packing factor, APF.

$r = 0.143\ nm = 0.143 \times 10^{-9}\ m$

1) $N = N_i + \frac{N_f}{2} + \frac{N_c}{8} = 0 + \frac{6}{2} + \frac{8}{8} = 3+1 = 4$

2) Edge length (x):
rf $d = 4r \Rightarrow (4r)^2 = x^2 + x^2$
$\Rightarrow x = 2\sqrt{2}r$

3) $APF: \frac{\text{Volume of atoms}}{\text{Volume of cube}} = \frac{4(\frac{4}{3}\pi r^2)}{x^3} = \frac{\frac{16}{3}\pi r^2}{(2\sqrt{2}r)^3} = 0.74$

   b. Calculate the theoretical density of the metal in the unit cell. How does this compare to the actual density of aluminium? Compare with the real-world density.

$\rho = \frac{n \cdot A}{V_c \cdot N_A}$, n = # atoms per unit cell (4) FCC

A: atomic weight $= 26.98\ g/mol$

V_c: unit cell volume $= x^3$

N_A: Avogadro's # $= 6.022 \times 10^{23}\ \frac{atoms}{mol}$

$r = 0.143\ nm$

$\Rightarrow x = 2\sqrt{2}(0.143) = 0.404\ nm = 0.404 \times 10^{-7}\ cm$

$V_c = x^3 = 6.60 \times 10^{-23}\ cm^3$

$\rho = \frac{4\ \cancel{atoms}(26.98\ \frac{g}{\cancel{mol}})}{6.60 \times 10^{-23}\ cm^3 (6.022 \times 10^{23}\ \frac{\cancel{atoms}}{\cancel{mol}})}$

$\Rightarrow \rho = 2.715\ \frac{g}{cm^3}$ calculated

real density: $2.7\ g/cm^3$

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4. You are a process engineering intern at a pharmaceutical firm developing a high-potency Active Pharmaceutical Ingredient (API) for pancreatic cancer treatment. During a routine characterization of the batch using X-ray diffraction (XRD) and microscopy, you identify significant **lattice defects** within the crystal samples.
   a. Should you be concerned about the presence of these defects regarding the final drug product? Justify your answer by discussing how defects can impact the **solubility** and **stability** of a pharmaceutical crystal. (4 points)

Defects in this case should be concerning because they can make the API's solubility and stability variable, which is risky for a final drug product.

Defects can affect solubility and stability in the following ways:

- Solubility: Defects make the crystal higher energy and less tightly held together, so drug molecules leave the crystal more easily. This can make the drug dissolve faster but inconsistently, leading to unpredictable solubility.
- Stability: Defects make the crystal less stable, which means the drug can change over time. The crystal may rearrange into a different form, slowly reorganize while in storage, absorb moisture more easily, or break down faster. These changes can alter how the drug dissolves and performs when it is used.

b. Identify **two** specific molecular or environmental factors during the crystallization process that could have led to these defects (4 points)

Two factors that could've led to these defects are:

- Rapid crystallization:
  - When crystals grow too quickly, molecules do not have enough time to arrange into an ordered lattice, resulting in vacancies, dislocations, and stacking faults.
- Impurities or solvent inclusion:
  - Trace impurities, residual solvents, or additives can become trapped within the growing crystal, disrupting the lattice and creating defects.

c. **Identify two industrial processes** in which crystal defects are critical to the product's function. (2 points)

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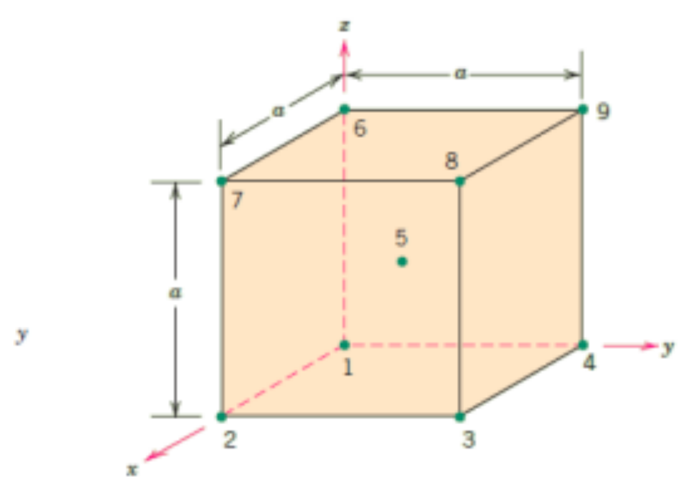
- Semiconductor doping in microelectronics:
  - Controlled crystal defects are necessary to tune electrical conductivity, enabling devices like transistors to function.
- Heterogeneous catalysis (industrial catalysts):
  - Crystal defects are required because reactions occur at defect sites (vacancies, dislocations). Without defects, catalytic activity would be much lower.

5. Write the coordinates for each of the points in the BCC (5 Points)



1) (0,0,0)
2) (1,0,0)
3) (1,1,0)
4) (0,1,0)
5) $(\frac{1}{2},\frac{1}{2},\frac{1}{2})$
6) (0,0,1)
7) (1,0,1)
8) (1,1,1)
9) (0,1,1)

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6. Calculate the equilibrium number of vacancies per cubic meter for silver at 800°C. The energy for vacancy formation is 1.10 eV/atom; the atomic weight and density (at 800°C) for silver are 107.9 g/mol and 9.5 g/cm³, respectively.(8 marks)
(Constants: Avogadro's number N_A = 6.022 x 10^{23} atoms/mol, Boltzmann's constant k = 8.617 x 10^{-5}eV/K).
Hint: **Refer to Chapter 4 of Materials Science and Engineering An Introduction by William D. Callister Jr. David G. Rethwish).**

$$N_v = N \exp\left(-\frac{Q_v}{kT}\right)$$

, N : # of atomic sites / m^3

$$N = \frac{\rho N_A}{A} = \frac{9.5\times10^6 \,g/m^3 \left(6.022\times10^{23} \frac{atoms}{mol}\right)}{107.9 \frac{g}{mol}}$$

$$N = 5.3\times10^{28} \frac{atoms}{m^3}$$

$$T = 800°C = 1073.15 \text{ K}$$

$$N_v = 5.3\times10^{28} \frac{\cancel{atoms}}{m^3} \exp\left(\frac{-1.10 \frac{\cancel{eV}}{\cancel{atom}}}{8.617\times10^{-5} \frac{\cancel{eV}}{\cancel{K}}(1073.15\ \cancel{K})}\right)$$

$$\boxed{N_v = 3.62\times10^{23} \frac{vacancies}{m^3}}$$

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7. For the purification of hydrogen gas by diffusion through a palladium sheet, compute the number of kilograms of hydrogen that pass per hour through a 5-mm-thick sheet of palladium having an area of 0.20 m² at 500°C. Assume a diffusion coefficient of 1.0 · 10^{-8} m²/s, that the concentrations at the high- and low-pressure sides of the plate are 2.4 and 0.6 kg of hydrogen per cubic meter of palladium, and that steady-state conditions have been attained. (10 points)

Fricks Law:

Flux

$$J = -D\frac{dC}{dx} \,[=]\, \frac{kg}{m^2 \cdot s}$$

, Area = 0.2 m^2
Δx = 5 mm = 5×10^{-3} m
$D = 1.0(10^{-8})\frac{m^2}{s}$
C_{High} = 2.4 kg/m^3
C_{low} = 0.6 kg/m^3

Flux (Area)

Diffusion: $J(A) [=] \frac{kg}{m^2 \cdot s}(m^2) = \frac{kg}{s}$

$$J = -1.0(10^{-8})\frac{m^2}{s}\left(\frac{(0.6-2.4)\frac{kg}{m^3}}{5(10^{-3})\,m}\right) = 3.6(10^{-6})\frac{kg}{m^2 \cdot s}(0.2\,m^2) = 7.2\times10^{-7}\frac{kg}{s}\left(\frac{3600\,s}{h}\right)$$

$$= 2.59\times10^{-3}\frac{kg}{h}$$

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8. An FCC iron-carbon alloy initially containing 0.20 wt% C is carburized at an elevated temperature and in an atmosphere in which the surface carbon concentration is maintained at 1.0 wt%. If, after 49.5h, the concentration of carbon is 0.35wt% at a position 4.0 mm below the surface, determine the diffusion coefficient and the temperature at which the temperature was carried out. (5 points + 5 extra credit)

?

a. Calculate the diffusion coefficient in this alloy under these conditions.

Frick's 2nd Law:

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

Given:

$C_0 = 0.20$ (initial carbon)
$C_s = 1.0$ (surface carbon)
$C_x = 0.35$ (at depth x)

need in same units wt.

$x = 4$ mm $= 4\times10^{-3}$ m

$t = 49.5$ h $= 1.782\times10^5$ s

$$\frac{0.35-1}{1-0.2} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

$$\Rightarrow 0.8125 = \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

from table:

$$\frac{x}{2\sqrt{Dt}} \approx 0.748 \quad 0.93$$

$$\Rightarrow 2\sqrt{Dt} = \frac{x}{0.748}$$

$$\Rightarrow \sqrt{Dt} = \frac{x}{2(0.748)}$$

$$D = \left(\frac{x}{2(0.748)}\right)^2 \cdot \frac{1}{t}$$

$$D = \left[\frac{4\times10^{-3}\,\text{m}}{2(0.748)}\right]^2 \cdot \frac{1}{1.782\times10^5\,\text{s}}$$

$$\boxed{D = 4.01\times10^{-11}\ \frac{m^2}{s}}$$

X

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b. Calculate the temperature (in Kelvin) at which this heat treatment was carried out (extra credit)

Assume the following constant for diffusion of carbon in γ-iron (FCC):

- Pre-exponential constant (D_o) = 2.3 x 10^{-5} m^2/s
- Activation energy (Q_d) = 148,000 J/mol
- Gas constant (R) = 8.314 J/(mol.K)

Note: you will require a Table of error function, which is attached along with the homework.

$$D = D_o \exp\left(-\frac{Q_d}{RT}\right)$$

$$\Rightarrow \frac{D}{D_o} = \exp\left(\frac{-Q_d}{RT}\right)$$

$$\Rightarrow \ln\left(\frac{D}{D_o}\right) = \frac{-Q_d}{RT}$$

$$\Rightarrow T = \frac{-Q_d}{R \cdot \ln\left(\frac{D}{D_o}\right)} = \frac{-148{,}000 \frac{\cancel{J}}{\cancel{mol}}}{8.314 \frac{\cancel{J}}{(\cancel{mol}\,K)} \cdot \ln\left(\frac{4.01 \times 10^{-11}}{2.3 \times 10^{-5}}\right)}$$

$$\boxed{T = 1342.52 \text{ K}}$$

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